Quadratic equations — factorising and the formula

Cambridge insert: the two methods, and knowing within five seconds which one this equation wants.

LESSON 2–3 | 2 × 40 MIN | ALGEBRA 10, REVISION | P1 · 1.1, 1.4–1.5

LESSON CLOCK

6' | 14' | 12' | 6' | 4'

Five equations — which method?	6 min
Factorising, including $a \neq 1$	14 min
The formula	12 min
Disguised quadratics	6 min
Homework	4 min

LEARNING OBJECTIVES

- Factorise a quadratic with leading coefficient 1 and with $a \neq 1$.
- Use the quadratic formula accurately.
- Choose between the two methods quickly.
- Solve equations that become quadratic after a substitution.

TEXTBOOK

National	Повторение, pp. 6–13
Cambridge	Pure Mathematics 1, pp. 2–14
Workbook	P1 Exercise 1A, 1D

01 Explanation

Two methods, one decision

THE ZERO PRODUCT RULE

If $A \cdot B = 0$ then $A = 0$ or $B = 0$. That is the whole reason factorising solves equations — and the reason the equation must be written with **zero on one side first**.

THE EQUATION LOOKS LIKE	USE
$x^2 + bx + c$ with small whole b, c	factorising — try it for five seconds
$ax^2 + bx + c$ with $a \neq 1$	factorising by splitting the middle term
ugly numbers, or the answer is asked for in surd form	the formula
no constant term	take out x — never divide by it

Factorising $2x^2 + 7x + 3$: find two numbers multiplying to $2 \cdot 3 = 6$ and adding to 7 — they are 1 and 6. Split the middle term and group:

$$2x^2 + x + 6x + 3 = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$$

The formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

It never fails, and it is slower. Use it when factorising has not worked in five seconds, or when the question asks for an exact answer.

THREE PLACES MARKS ARE LOST

1. The whole of $-b$ is over the $2a$, not just the root.
2. $-b$ where b is already negative gives a **plus**.
3. $2a$, not 2, when $a \neq 1$.

Equations that become quadratic

Many equations are quadratics wearing a disguise. Substitute a letter for the repeated block and the disguise falls off:

$$x^4 - 5x^2 + 4 = 0 \ \text{ let } t = x^2 \implies t^2 - 5t + 4 = 0$$

Solve for t, then go back: $t = 1$ gives $x = \pm 1$, $t = 4$ gives $x = \pm 2$. Four roots.

The same trick handles 2^x, $\sqrt{x}$ and $\frac{1}{x}$ as the repeated block — and each time, **check the substitution allows the value** before converting back.

02 Worked examples

EXAMPLE 1 **Solve $3x^2 - 10x + 8 = 0$ by factorising.**

1. $3 \cdot 8 = 24$, and $-4 + (-6) = -10$

   Two numbers multiplying to 24, adding to −10.

2. $3x^2 - 4x - 6x + 8 = 0$

   Split the middle term.

3. $x(3x - 4) - 2(3x - 4) = 0$

   Group.

4. $(3x - 4)(x - 2) = 0$

5. $x = \frac{4}{3}$ or $x = 2$

ANSWER

$x = \frac{4}{3}$, $x = 2$

EXAMPLE 2 **Solve $2x^2 - 6x + 1 = 0$, giving exact answers.**

1. $a = 2, b = -6, c = 1$

2. $D = 36 - 8 = 28$

3. $$x = \frac{6 \pm \sqrt{28}}{4}$$

   Note $-b = +6$.

4. $\sqrt{28} = 2\sqrt{7}$

5. $$x = \frac{6 \pm 2\sqrt{7}}{4} = \frac{3 \pm \sqrt{7}}{2}$$

   Cancel the 2.

ANSWER

$$x = \frac{3 \pm \sqrt{7}}{2}$$

EXAMPLE 3 **Solve $x^4 - 13x^2 + 36 = 0$.**

1. $t = x^2$

   and $t \geq 0$.

2. $t^2 - 13t + 36 = 0$

3. $(t - 4)(t - 9) = 0$

4. $t = 4$ or $t = 9$

   Both are ≥ 0 ✓

5. $x = \pm 2$ or $x = \pm 3$

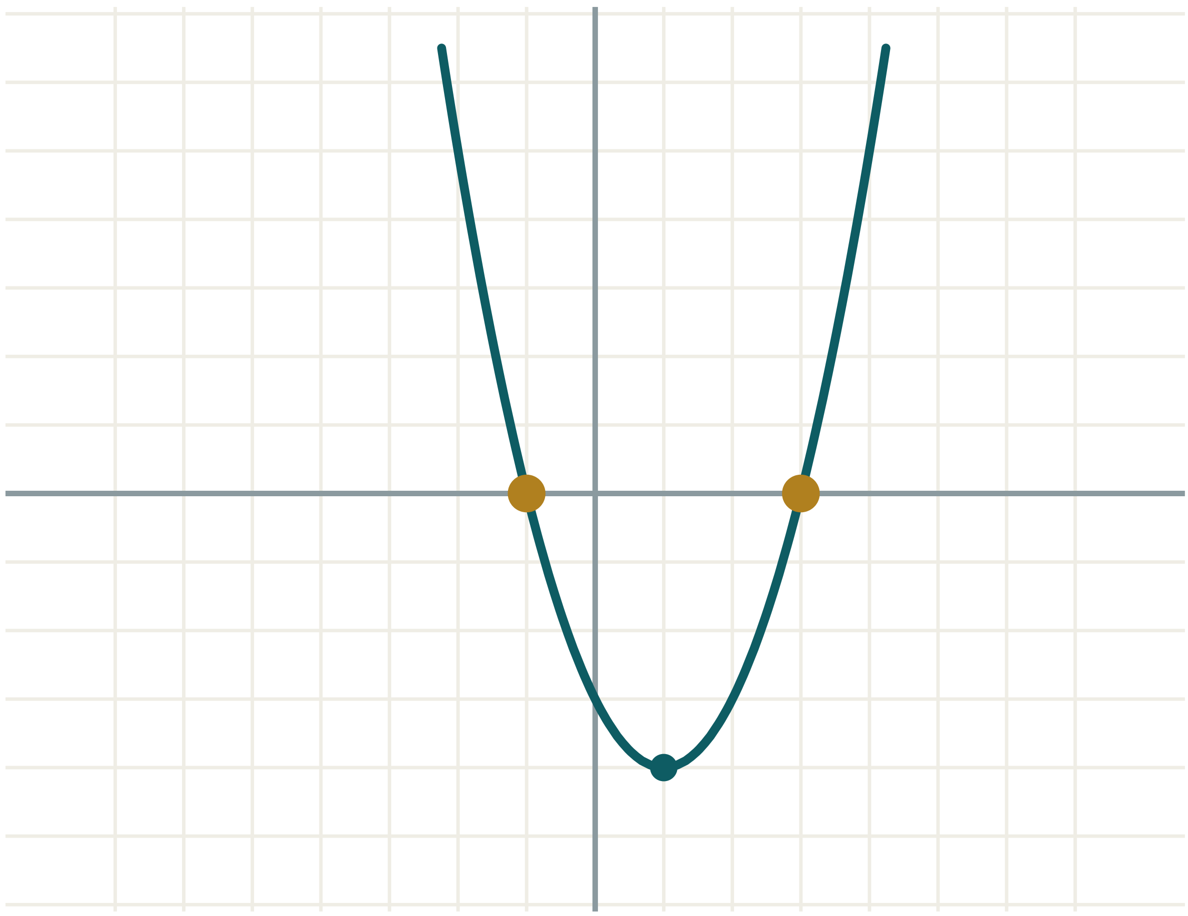
ANSWER

$x = \pm 2, \pm 3$

03 Interactive model

Slide the coefficients and watch the roots appear and disappear as the parabola lifts off the axis.

The roots of a quadratic



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | x_1 **-1** | x_2 **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Coefficient	Koeffitsient	Коэффициент
Factorise	Ko‘paytuvchilarga ajratish	Разложить на множители
Root	Ildiz	Корень
Quadratic formula	Kvadrat tenglama formulasi	Формула корней
Zero product rule	Nolga ko‘paytma qoidasi	Правило нулевого произведения
Substitution	O‘rniga qo‘yish	Подстановка
Disguised quadratic	Yashirin kvadrat tenglama	Уравнение, сводящееся к квадратному
Exact value	Aniq qiymat	Точное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$x^2 - x - 6 = 0$ has roots:

- $2, -3$
- $3, -2$
- $6, -1$
- $1, -6$

For $2x^2 + 3x - 2 = 0$ the formula gives $2a$ equal to:

- 2
- 4
- 3
- -2

$x^2 = 7x$ has roots:

- 7
- $0, 7$
- $\pm\sqrt{7}$
- 0

In $x^4 - 5x^2 + 4 = 0$, substituting $t = x^2$ gives how many values of x?

- 2
- 3
- 4
- 1

Which method suits $x^2 - 4x + 1 = 0$?

- factorising
- the formula
- take out x
- no method works

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x^2 - 5x + 4 = 0$

   ANSWER $x = 1, x = 4$

2. Solve $x^2 + 7x + 10 = 0$

   ANSWER $x = -2, x = -5$

3. Solve $x^2 - 25 = 0$

   ANSWER $x = \pm 5$

4. Solve $x^2 + 3x = 0$

   ANSWER $x = 0, x = -3$

5. Solve $2x^2 - 8 = 0$

   ANSWER $x = \pm 2$

6. Solve $x^2 - 2x + 1 = 0$

   ANSWER $x = 1$ (repeated)

7. Solve $x^2 + x - 20 = 0$

   ANSWER $x = 4, x = -5$

Medium · the standard question

7 PROBLEMS

1. *Solve* $2x^2 + 5x + 2 = 0$

   ANSWER $x = -\frac{1}{2}, x = -2$

2. *Solve* $3x^2 - 7x + 2 = 0$

   ANSWER $x = 2, x = \frac{1}{3}$

3. *Solve* $x^2 - 4x + 2 = 0$ *exactly.*

   ANSWER $x = 2 \pm \sqrt{2}$

4. *Solve* $5x^2 - 3x = 0$

   ANSWER $x = 0, x = 0.6$

5. *Solve* $x^4 - 10x^2 + 9 = 0$

   ANSWER $x = \pm 1, \pm 3$

6. *Solve* $4x^2 - 12x + 9 = 0$

   ANSWER $x = 1.5$ (repeated)

7. *Solve* $6x^2 - 5x - 6 = 0$

   ANSWER $x = \frac{3}{2}, x = -\frac{2}{3}$

Hard · stretch and reason

7 PROBLEMS

1. *Solve $3x^2 + 2x - 4 = 0$ exactly.*

   ANSWER $x = \dfrac{-1 \pm \sqrt{13}}{3}$

2. *Solve $x^4 - 3x^2 - 4 = 0$*

   ANSWER $t = 4$ only (reject $t = -1$), so $x = \pm 2$

3. *Solve $2^{2x} - 6 \cdot 2^x + 8 = 0$*

   ANSWER $2^x = 2$ or 4, so $x = 1, 2$

4. *Solve $x - 5\sqrt{x} + 6 = 0$*

   ANSWER $\sqrt{x} = 2$ or 3, so $x = 4, 9$

5. *The roots of $x^2 + px + 12 = 0$ differ by 1. Find p.*

   ANSWER $p = \pm 7$

6. *Solve $\dfrac{1}{x^2} - \dfrac{5}{x} + 6 = 0$*

   ANSWER $t = \dfrac{1}{x}$: $t = 2, 3$, so $x = \dfrac{1}{2}, \dfrac{1}{3}$

7. *Show that $x^2 + x + 1 = 0$ has no real root, without the formula.*

   ANSWER $(x + \dfrac{1}{2})^2 + \dfrac{3}{4}$ is a square plus a positive number, so never zero.

07 Homework

Homework — 6 problems

P1 Exercise 1A and 1D. Say which method you chose and why.

1. *Solve $x^2 - 8x + 15 = 0$*
2. *Solve $2x^2 - 7x + 3 = 0$*
3. *Solve $x^2 - 6x + 4 = 0$ exactly.*

4. *Solve* $3x^2 + 5x = 0$

5. *Solve* $x^4 - 5x^2 + 4 = 0$

6. *Solve* $x - 7\sqrt{x} + 12 = 0$